Pakhi Banchalia

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Research Experience

- **Computational Machinery of Cognition (CMC) lab, TU Dresden** — *Nov 2024 - Present*
  Research Intern — Dresden, Germany
  - Designed homeostatic reinforcement learning agents to study task generalization and retention under internal drive constraints
  - Implemented and evaluated Dopamine, a gradient-free learning algorithm based on reward-modulated plasticity
- **Lossfunk** — *July 2025 - Sept 2025*
  Research Intern — Bangalore, India
  - Investigated homeostatic learning in the Crafter environment
  - Analyzed agent behavior under varying internal reward structures and compared against baseline approaches
- **Google Summer of Code, INCF** — *May 2024 - Sept 2024*
  Contributor — Remote
  - Simulated *C. elegans* neural development using Neural Cellular Automata and evolutionary optimization
  - Modeled neural connectivity with graph neural networks
  - Applied Kepler Mapper (topological data analysis) to analyze developmental trajectories
  - Produced open-source, reproducible research code

Publications

- Pillai, H.S., **Banchalia, P.**, Jamboji, V., Krishnan, S.R. (2026). *CAN: Continuously Adapting Network*. In *Data Science and Applications*, ICDSA 2025. Lecture Notes in Networks and Systems, vol 1723. Springer, Cham. [DOI]

Research Projects

- **Homeostatic Crafter**
  - Explored Homeostatic Reinforcement Learning in Craftax environment
  - Observed agent behaviour in different reward settings
  - Tried introducing modularity to further improve agent performance
- **Dopamine: Reward-Modulated Gradient-Free Learning**
  - Contributed to a biologically inspired learning rule using reward-modulated weight perturbation
  - Explores credit assignment without backpropagation
  - Achieved competitive performance in preliminary benchmarks
- **SEANN : Self-Expanding and Adaptive Neural Networks**
  - Proposed a neural architecture that dynamically grows capacity for continual learning
  - Identifies task-relevant subnetworks to reduce catastrophic forgetting

Technical Skills

- **Languages:** Python, C++, MATLAB, SQL
- **ML / Research:** PyTorch, TorchVision, Gymnasium (OpenAI Gym), JAX (familiar)
- **Scientific Computing:** NumPy, SciPy, Pandas, Matplotlib, Kepler Mapper (TDA)
- **Research Areas:** Reinforcement Learning, Biologically-Inspired Learning, Neural Cellular Automata, Continual Learning, GNNs
- **Developer Tools:** Git, Linux/Unix, Docker, LaTeX, Weights Biases (WB)

EDUCATION

- **Technische Universität Dresden** *Oct 2025 - Present*
  Master of Science in Computational Modeling and Simulation (Applied AI track) Dresden, Germany
  - GPA: not yet available
- **Amrita Vishwa Vidyapeetham Kerala, India** *2021-2025*
  Bachelor of Technology in Computer Science and Artificial Intelligence Amritapuri, India
  - Grade: 8.8/10